BAMS
Meeting Summary

Hot Topics: Building Capacity for Extreme Heat Adaptation through Transdisciplinary Training

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Adaptation;
Health impacts;
Heat wave;
Climate;
Education;
Vulnerability

The Inaugural Global School on Heat Adaptation

What: The Global School on Heat Adaptation is a weeklong workshop-style educational opportunity focused on transdisciplinary learning and training of the next generation of heat and health researchers and practitioners to address the growing challenges of extreme heat. The school comprised 24 international graduate students and was guided by the in-depth participation of 13 researchers and partners. In total, 10 countries and 15 disciplines were represented. Topics ranged from active learning sessions focused on physiological responses at the microscale to personal-level heat exposures and behaviors, up to neighborhood-, city-, and county-level actions. Activities hosted throughout the week integrated elements of policy, climate variability and change, cultural contexts, and the built environment.

When: 21–25 July 2025

Where: Montreal, Quebec, Canada

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1. Introduction: Context and purpose

Extreme heat is a global challenge that requires local-to-personal-level adaptation strategies and solutions to mitigate adverse impacts on human health, well-being, and productivity (Ebi et al. 2021). Significant gaps in knowledge and practice remain, particularly in quantifying heat exposure, understanding the subsequent physiological and behavioral responses, and developing nonstandard methods to study human adaptation, which limit our ability to synthesize research and thrive in a hotter climate (Vecellio and Vanos 2024). To enhance the practice and study of these complex topics, gaps in training and education must be addressed. Postsecondary education is still primarily delivered in disciplinary silos, with minimal stakeholder engagement. Given the convergent nature of heat challenges and solutions, this siloed approach does not equip students with the skills, knowledge, and experiences necessary to holistically understand heat as a multifaceted, complex challenge encompassing physiological, behavioral, infrastructural, and societal elements. This fragmentation across disparate disciplines limits the creativity of individuals and organizations, impedes innovation, and slows the translation of concrete solutions into action for end users.

To address this challenge, our team of researchers and practitioners at the Global Center for Heat Adaptation (https://heatadaptation.asu.edu/) created and delivered the first-ever weeklong Global School on Heat Adaptation (https://icm-mhi.org/etudiant/global-school-on-heat-adaptation/), hosted at the Montreal Heart Institute. Both initiatives were funded by the National Science Foundation's Global Centers program, in partnership with the Social Sciences and Humanities Research Council of Canada. The school ambitiously engaged graduate students, researchers, and stakeholders in active experiential learning and cross-disciplinary exchange to foster the creation of more use-inspired, real-world solutions. In this meeting summary, we provide an overview of attendees and activities, offer insights into best practices, share participant experiences and outcomes, and outline plans for future iterations of the Global School on Heat Adaptation.

2. Attendees

The opportunity was advertised globally on multiple websites and platforms to attract a diverse group of graduate students from different disciplines. For this inaugural edition, 20 graduate students—six at the master's level and 14 at the doctoral level—were selected for

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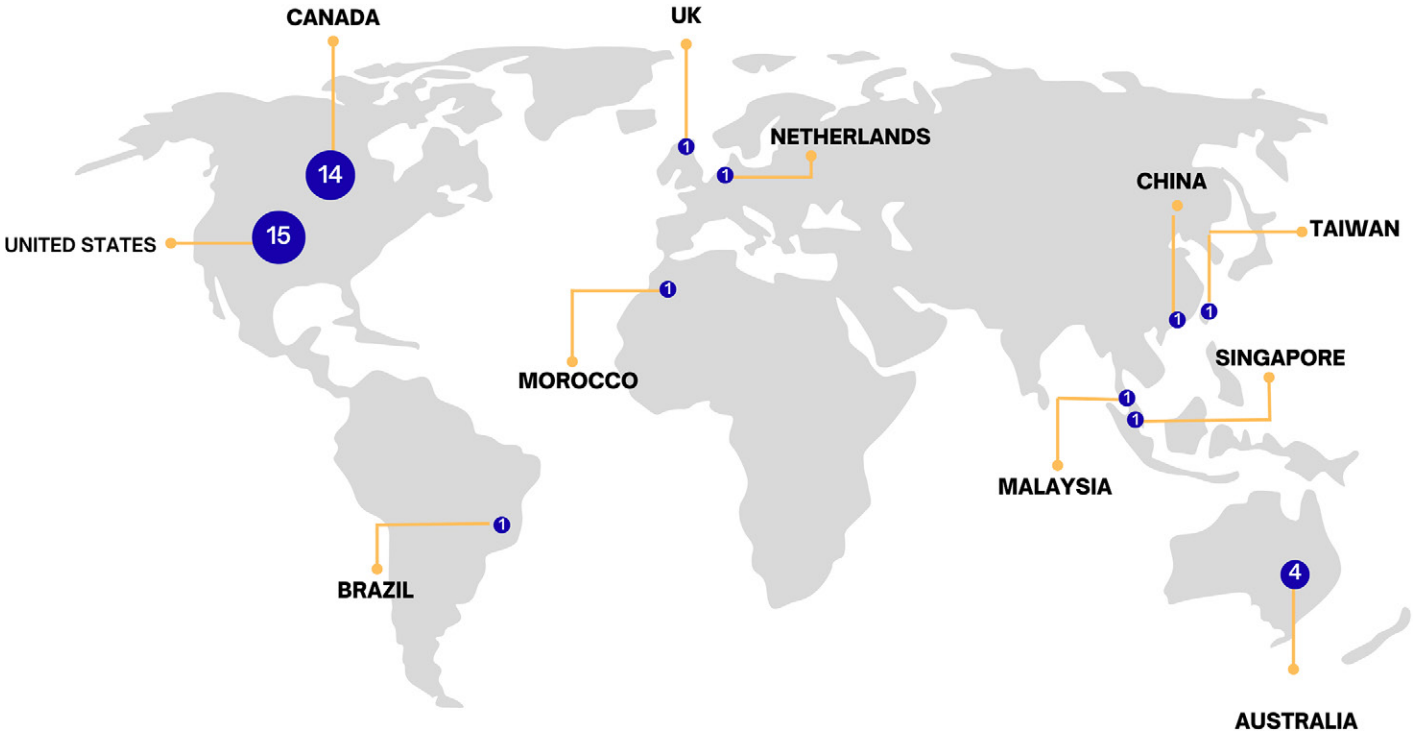


Fig. 1. Simple map overview of the locations of home institutions for all participants, including graduate students, facilitators, and stakeholders. Each circle indicates the number of participants from the given country. While this map depicts the location of study or work for each participant, there were 13 countries represented based on participants' home location, indicating diverse international representation.

participation from a pool of over 50 applicants from 21 countries. The selection process was competitive and aimed to ensure balanced representation across study levels, gender, disciplines, and countries of origin. Student backgrounds spanned physiology, medical sciences, landscape architecture, policy, climate science, psychology, and sports science. Participants brought global perspectives from both their home countries and countries of study (Fig. 1), with the majority traveling from universities within North America, Europe, and Australia. Travel bursaries that covered all expenses to participate in the school were provided to students from lower- to upper–middle-income countries, allowing representation from Malaysia, Brazil, China, and Morocco. The course was organized and run by researchers, practitioners, and trainees affiliated with the Global Center for Heat Adaptation. The local team at the Montreal Heart Institute provided on-site logistical support, food, guidance, and space. When combining the selected students with trainees (students and postdocs) from teams affiliated with the Global Center for Heat Adaptation, a total of 28 learners participated, including 16 females and 12 males.

3. Activities

The weeklong school in Montreal was designed using thoughtful, innovative, and use-inspired approaches to ensure that novel teaching and learning methods led to deeper understanding in a safe and fun environment. A summary of the session topics and learning objectives is provided in Table 1. Strategies included active learning with meteorological and physiological sensors, a heat chamber, and a thermal suit; guided tours; rich discussions; collaborative problem-solving exercises; and gamified learning (see images in Fig. 2). Each day was organized around a central theme.

Prior to the workshop, efforts were undertaken to create excitement and cohesion among participants. Summary sheets with speaker bios and learning objectives were distributed for each session, along with suggested reading materials. Surveys were sent to participants to

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Table 1. Session topics and activities across diverse disciplines, taught by both disciplinary experts as well as practitioners, often at the city or local levels. The diversity of sessions and mixed approaches to learning was designed to promote experiential learning, problem-solving, open discussion, critical reflection, and storytelling to achieve greater impact. All sessions left time for discussion across the whole group, at each table, and Q&A. Further information is provided within the Open Science Framework repository (https://doi.org/10.17605/OSF.IO/MU73Z).

Topic and methods of teaching	Learning objectives
Mixed-method teaching and learning	
Global center and school overview Presentations and group introductions	• Provide an overview of the creation, purpose, vision, and team behind the Global Center for Heat Adaptation Track-2 design grant. • Differentiate the project thrusts (collaboration, learning, research). • Overview of the primary research tool in the project: The HeatSuite. • Outline past efforts (e.g., Phoenix workshop, HeatSuite research) and future plans.
Climate change and heat attribution Direct instruction and discussion	• Explain the physical basis of climate change (including natural- or climate-induced impacts), common misconceptions, and important atmospheric data products. • Understand the process and purpose of climate change projections and apply them to heat-health outcomes. • Review climate terms commonly used in heat-health literature (e.g., RCP–SSP)[a]. • Understand the human cost of heat and mortality statistics and interpret the attribution of a heat death to climate change.
Urban climate and heat Direct instruction and hands-on sensor activities	• Understand how urban form, design, and landscaping influence local microclimates and human thermal exposure. • Differentiate between heat metrics [air, surface, and mean radiant temperature, wet-bulb globe temperature (WBGT), etc.] and which best reflect the human heat experience. • Explore how the built environment contributes to and/or mitigates urban heat through land cover, materials, and form, using case studies from Phoenix, Arizona.
HeatReady neighborhoods and schools Storytelling, direct instruction, real-world examples	• Examine the differing narratives of extreme heat through storytelling. • Explore how personal, social, and structural factors influence heat vulnerability and resilience • Reflect on one's own experiences, applying a "heat lens" to work • Explore the HeatReady neighborhoods and HeatReady schools frameworks as tools for advancing thermal justice.
Intersectoral adaptation Direct instruction, real-world examples	• Share pan-Canadian guidance and practical actions that can be implemented to reduce extreme heat risk for various stakeholder groups. • Showcase approaches for motivating behavior change and encouraging adoption of protective actions for extreme heat (and other extreme weather events).
Pathophysiology of heat-related health risks Direct instruction, discussion based	• Distinguish between heat stress and heat strain. • Explain the main physiological responses during heat exposure. • Understand how physiological strain underlies heat-related health impacts.
Technologies for in situ heat stress and strain Direct instruction	• Identify strengths and limitations of current commercial wearable devices for monitoring heat health. • Establish links between key biomarkers and homeostatic balance. • Explain how technology may improve heat resilience.
Behavioral responses in the heat Direct instruction and group activity	• Distinguish between autonomic and behavioral responses to heat. • Explain the importance of thermal discomfort in cool-seeking behavior. • Describe the factors influencing thermal discomfort during heat exposure. • Critically think about the interaction between behavior and physiology during real-world scenarios.
Resources and tools for motivating heat adaptation Direct instruction and scenario-based learning	• Gain an understanding of practical solution sets for adaptation. • Develop an understanding of how residents and communities can be motivated to change behavior in the direction of enhanced protection for extreme heat (and other weather hazards). • Take on the role of "chief resilience officer" for a city and complete a scoring tool to determine how prepared the city is for managing extreme heat risk.
Montreal heat response plan Direct instruction and group activity	• Share the Montreal experience related to implementation of its heat response plan over the past 20 years, with an accent on continuous improvement and innovation, intersectoral collaborations, and citizen engagement. • Mobilize the collective intelligence of the group around issues in Montreal related to the heat response plan.

(*Continued*)

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Table 1. (*Continued*).

Topic and methods of teaching	Learning objectives
Heat stress during sport: From policy to practice Direct instruction, group problem-solving, discussion	• Equip participants with knowledge and practical tools to effectively manage heat stress during physical activity and sport (elite and recreational settings). • Explore the physiological principles of thermoregulation; evidence-based prevention and mitigation strategies; and best practices to safeguard performance, health, and safety in hot environments. • Introduce additional physiological and practical considerations for protecting children during physical activity in high-temperature conditions.
Active sharing and experiential learning	
Climate Fresk Active learning, discussions, problem-solving	• Enable an open, positive conversation about climate change mechanisms, challenges, and solutions. • Raise awareness and understanding of climate change through an interactive, collaborative, and creative workshop. • Explore the causes, consequences, and interconnections of climate change to foster critical thinking and discussion about solutions. • Blend expertise from various disciplines to solve complex climate challenges. • Facilitate camaraderie among all participants (students and facilitators).
Poster presentations	• Present proposed or ongoing research related to heat adaptation. • Gain feedback from all participants in various disciplines related to extreme heat and health.
Experiential learning: 1) Thermal chamber 2) Water perfused suit 3) HeatSuite demo Hands-on activities using different modalities of heat exposure and measurements	• Illustrate concrete examples on the use of a heat chamber and a thermal suit using real subjects, demonstrating live data collection using multiple devices, sensors, and measurement platforms. Real-time demonstrations and Q&A allowed for consolidation of learning. • Enable students to use a multimodal remote data collection system (e.g., HeatSuite) to collect, transmit, and visualize environmental, physiological, and behavioral data from participants' homes. • Explore potential applications in one's own research based on new knowledge of the three methods, techniques learned throughout, and the numerous devices available. • Gain feedback from students and experts to improve the current equipment.
Knowledge consolidation activity Collaborative group work, solutions focused	• Consolidate learning that occurred during the week. • Collaborate to frame new learnings into a proposal idea that focuses on heat and health solutions. • Articulate ideas in a short team presentation.
Cultural experiences, group socializing, and relationship building	
Throughout the week, all participants had opportunities to partake in networking events (group dinners, hikes, local activities, group runs, or walks) based on personal interest. These activities promoted group cohesion and community building. A particular highlight was a hike up Mount Royal, a mountain in the City of Montreal with deep historical roots tied to Indigenous and European populations. It serves as a central hub for religious, educational, and recreational activities, offering citywide views.	

[a] Representative concentration pathway–shared socioeconomic pathway.

gauge their interests, expectations, and preferences, which were used to inform the development of the final program, social activities, and meals. A practical guide to Montreal was provided to help participants navigate the city, find lodging, and explore attractions. Participants also shared brief "get to know you" bios, and a group was created on a free global messaging platform to connect participants, facilitate communication, and address shared expenses (e.g., accommodation).

The first day was organized around the theme of climate change, a scientific topic familiar to some disciplines but newer to others. Background on the science of climate change and heat-health attribution was provided. This session was followed by an interactive "ice-breaker" activity for all students and facilitators: the "Climate Fresk" workshop (https://climatefresk.org/world/), animated by a facilitator from the Climate Fresk association. The workshop included hands-on and group-based activities, discussions, and explanations to understand the mechanisms of climate change and to foster an open, positive dialogue about solutions.

Day 2 centered on the theme of urban heat, discussing the built environment and microscale impacts on "heat" (including air, surface, and radiant temperatures), various measurement

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Fig. 2. Images across various activities and sessions at the weeklong Global School on Heat Adaptation. (a) Full group photo of participants and facilitators; (b),(c) Climate Fresk activity, working through and understanding the "climate puzzle" using cards with imagery to delineate relationships between the causes, consequences, and solutions to climate change; (d),(e) thermal suit experiential learning where participants were exposed to tools and techniques for using a water perfused suit for mechanistic research to help understand physiological responses to heat stress; (f) thermal chamber experiential learning activity, where students entered a thermal chamber (set to 38°C air temperature and 50% relative humidity) to observe the various instruments and techniques used in common thermal chamber studies; (g) "Hackathon," where participants collaborated to frame new learnings into a grant proposal idea focused on heat and health solutions; (h) HeatSuite demonstration, where participants were able to obtain data such as blood pressure, heart rate, temperature, and body metrics (height, weight) and observe how they are automatically transferred via Bluetooth to a main node for field based research; and (i) heat stress in sports problem-solving activity, focused on developing a method to protect athletes from the heat in real-world scenarios.

tools, and real-world studies of urban heat mitigation strategies. Social vulnerability and heat resilience were also addressed through storytelling, using HeatReady neighborhoods and schools as examples (Shortridge et al. 2022; Guardaro et al. 2025). Finally, locally based collaborative climate resilience strategies were presented, with an emphasis on intersectoral collaboration and culturally relevant barriers. The day culminated in student poster presentations and lightning talks on their research, fostering socializing, networking, and feedback in a comfortable, friendly environment.

Day 3 focused on the physiology of heat exposure, building on previous days' learnings and unique vulnerabilities to ensuing health outcomes across diverse populations. Sessions explored how adverse heat impacts are ultimately caused by the body's physiological response to heat exposure (Gagnon et al. 2026), as well as technological challenges and opportunities for measuring heat exposure and physiological responses in daily life. The afternoon featured experiential learning, achieved by having small groups rotate through three stations: 1) simulating hot and humid extremes in an environmental chamber; 2) measuring physiological responses using a thermal suit; and 3) demonstrating the HeatSuite system to quantify environmental, behavioral, and physiological responses in real-world settings (Ravanelli et al. 2025) (see Fig. 2).

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Day 4 was dedicated to behavioral responses to heat, discussing many of the physiological signals and situational factors that drive or modify decisions to seek cooling (Schlader and Vargas 2019), and the importance of studying adaptive behaviors in field-based settings to inform policy and public health measures (Vargas et al. 2023). Students were also introduced to several practical approaches, tools, and resources for changing behavior and motivating adaptation to extreme heat, drawing lessons from other extreme-weather hazards (e.g., flooding, wildfires). In the afternoon, attendees had free time to work, discuss, gain feedback, and collaborate on their various research projects.

The final day focused on the public health response to extreme heat, with specific examples from the Montreal heat response plan and the development of the extreme heat policies of the Australian Open and Sports Medicine Australia (Tartarini et al. 2025). Two members of Montreal Public Health's citizen committee participated in the session, bringing their experience and expertise to the table. Participants worked in small groups to explore the essential contribution of advocacy in developing new approaches to community resilience in the context of extreme heat. All participants were also given the opportunity to develop their own heat policies for sports in a group setting.

In the afternoon, a group-based workshop was held to consolidate the week's learning. Small groups were formed, and each group was tasked with drafting a "front page" for a grant funding proposal, with the requirement that the proposal focuses on solutions, integrates across multiple disciplines, and involves stakeholders and/or partners. Once each team had drafted its front page, a representative from each team presented it to the larger group, and discussions ensued.

4. Evaluation and outcomes

At the conclusion of the weeklong event, the University Office of Evaluation and Educational Effectiveness (UOEEE) (https://uoeee.asu.edu/) at Arizona State University administered an anonymous survey to all student participants. The surveys prompted participants to reflect upon their experience and provide feedback. Twenty three of 26 students (88%) completed the survey. Below, we present information from the survey to help us understand students' experiences as reflected in their feedback. This information supports the next iteration of the Global School on Heat Adaptation.

According to the UOEEE report, student survey results indicated satisfaction with the Global School on Heat Adaptation Program in Montreal. The student participants felt that the event positively impacted them by deepening their awareness of societal challenges posed by extreme heat and by improving their knowledge of cutting-edge research on this topic. The most impactful aspects of the event included exposure to diverse perspectives and collaboration and practical and applied learning, as well as the program design and learning. Students frequently highlighted their positive experiences of learning from diverse global contexts and often disparate disciplines:

> The opportunity to gain diverse perspectives from colleagues and professionals on the complex challenges of heat from the climate side to human physiology, technological applications, social engagement and policy. The practical sessions were very relevant and informative. I also made connections with other students from across from diverse fields related to solving the issues of heatwaves globally.

Furthermore, 91% strongly agreed or agreed that the event deepened their awareness of everyday societal challenges posed by extreme heat and enhanced their understanding of cutting-edge research endeavors on extreme heat.

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The educational workshop also strengthened students' understanding and confidence in research approaches and collaboration with stakeholders. From the survey, 86% of students strongly agreed or agreed that the event prompted them to think critically about their own research methodologies and spurred new research interests/methodologies they would like to learn more about. Two relevant comments around the most impactful aspects include the following:

> Learning from the variety of multifaceted research examples that are tackling the challenge of heat adaptation and making connections with the leaders and participants.

> I found the group of people coming from differing disciplines and countries/contexts was most impactful. I think I must have mentioned every day that, "This is so cool!" or "I'm learning so much about physiology!" That would not have been possible without a diverse group of people coming together.

A majority of the students appreciated that the sessions were interactive and included opportunities to learn, collaborate, and network with other participants. They also liked the discussions generated from presentations. Participants highly rated various aspects of the learning, including the quality of the sessions they attended, the overall timing and scheduling of the event, and the pacing of the activities, though a few students offered constructive feedback on these topics. For example, one student commented as follows:

> For me, all the sessions are impactful, particularly the group problem-solving sessions, practical sessions on heat physiology, and the hackathon.

Students most frequently commented that developing research design and innovation and integrating physiological measurements and laboratory methods were among the aspects of the event they planned to apply in their own research methods. One participant commented:

> From the physiology lectures, I gained more understanding of the basic underlying science, which then in turn gave me more ideas to apply to my urban climate research.

Overall, the experience was beneficial for the students, with positive feedback, such as their appreciation for the event. One student commented as follows:

> Attending the Global School on Heat Adaptation was my best academic and professional decision for 2025.

Students have stayed connected through their built networks and the information shared, such as their initial biographies circulated before the meeting. Finally, the school attracted media attention, including three TV reports, six live radio interviews, and two live TV interviews across English and French languages, and over 20 written articles that showcased the initiative. Select media outputs include CityNews Montreal, CTV News, CBC Radio, and The Suburban.

5. Next steps

Students and facilitators expressed strong interest in holding additional iterations of the Global School on Heat Adaptation, aiming to hold the school annually or biannually in new locations worldwide and to incorporate emerging research. Final recommendations from the

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UOEEE evaluation for future iterations of the Global School on Heat Adaptation include the following:

- Continue collecting participants' feedback through surveys, interviews, or focus groups for future events.
- Continue providing opportunities for interactive learning, engaging collaborations, and networking among participants in future events; limit or remove virtual presentations.
- If possible, include more diverse content (e.g., epidemiological aspects/study of extreme heat and community-building strategies).
- Consider amending the Climate Fresk session (e.g., allow more time for this activity, revise the methodology to be more flexible, include an overview lecture on climate feedbacks, or target the Fresk more on extreme heat).
- Increase representation from additional countries.
- Include more community-focused sessions and/or stakeholder and partner engagement activities.

In summary, the inaugural Global School on Heat Adaptation brought together a diverse group of graduate students, researchers, practitioners, and stakeholders from across the world to engage in collaborative learning of the complex and multifaceted challenges posed by extreme heat. Through active, multiscale learning—from physiological responses to community-level actions—the program connected science, policy, climate change, culture, and the built environment to advance holistic approaches to extreme heat adaptation. Designed to train the next generation of heat and health researchers and practitioners, this initiative provides a concrete example of breaking down silos to integrate the multiple disciplines, sectors, and perspectives needed to identify and mobilize effective strategies to minimize the wide-ranging impacts of extreme heat on human health, well-being, and performance.

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Data availability statement. Materials, including the agenda, presentations, and posters, can be found in the Open Science Framework repository at https://doi.org/10.17605/OSF.IO/MU73Z.

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